

Database Users and Administrators

Types of Users, Roles, and Responsibilities

1.8 Database Users and Administrators

Definition

Database users are individuals who interact with a database system to store, retrieve, or manipulate data. Database administrators (DBAs) manage, maintain, and secure the database system.

1.8.1 Database Users

Users are categorized based on how they interact with the system:

1. Naive Users:

- Unsophisticated users who interact through pre-written application programs.
- Examples: web users, bank tellers, clerical staff.
- Interface: Forms-based or pre-generated reports.

2. Application Programmers:

- Professionals who write application programs.
- Use **Rapid Application Development (RAD)** tools for minimal coding effort.

3. Sophisticated Users:

- Form queries directly in a database query language.
- Examples: Analysts performing OLAP (Online Analytical Processing), data mining.

1.8.2 Database Administrators (DBA)

A **DBA** has central control over data and programs accessing the database. Main functions include:

1. Schema Definition:

- Creating the database schema using DDL statements.

2. Storage Structure and Access Method Definition:

- Define file organization (sequential, heap, hash, B+ tree).
- Define record organization (fixed/variable length).
- Define indexes (ordered index, hash index).

3. Schema and Physical Organization Modification:

- Update schema and physical storage to reflect changing requirements or improve performance.

4. Granting Authorization for Data Access:

- Regulate which users can access which parts of the database.

5. Specifying Integrity Constraints:

- Implement primary keys, foreign keys, triggers, assertions, and business rules.

6. Routine Maintenance:

- Periodic backup to tapes or remote servers to prevent data loss.
- Ensure adequate disk space and upgrade as needed.
- Monitor running jobs and database performance.

Note

DBAs are crucial for maintaining **data consistency, security, and availability** across the organization.